Section - A

Que-1
Ans 3 (2)

Que-2
Ans Femur (2)

Que-3
Ans Closing of Suture (2)

Que-4
Ans Male skull (2)

Que-5
Ans Identifying individuals (2)

Que-6
Ans Head shape (2)

Que-7
Ans Fingerprint (1)

Que-8
Ans All of the above (2)

Que-9
Ans Age at death (2)

Que-10
Ans They are Unique (2) to each individual

Section-B

Que-11. 2
Ans - Significance of Anthropology :- (6)

i) Anthropology means study the human behaviour, their cultural, societies from their skeletal remains. It helps in the forensic investigation to determine the background, state of the person.

ii) It helps to differentiate between the human and non-human from the skeletal remains which helps in solving the case.

iii) Study the anthropology helps to determine that it is human skull or animal skull.

iv) Forensic Anthropology helps to solve cases like mass disasters, tsunamis etc. to determine is it human or not?

v) Forensic anthropology also helps to determine that what is the reason of the deaths of the individual.

vi) It also helps to differentiate between the male and female skeletal system by their characteristic features.

vii) Anthropology also helps to determine the age from skull, sex from skull etc. which helps the forensic investigators to solve the case.

Forensic anthropology plays a crucial role in forensic science helps to determine the background and cultural of an individual.

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Que - 12.1

Ans Male skull -

- Male skull and female skull their characteristics features are different from each other.
- It helps to determine that the skull is belong to male or female.
- It helps the forensic investigators to solve the case.
- For an example, General appearance, frontal eminences, cheek shapes, chin shape, Architecture, Gonial angle, size of the skull etc. are different from the female skull.

There are some characteristics features of the male skull -

	Features	Male skull
i)	General appearence	Large
ii)	Architecture	Rugged
iii)	Orbits	Round with sharp Margins
iv)	Gonial angle	Angled
v)	Cheek shape	Broad
vi)	Robusticity	High pronounced
vii)	Frontal eminencies	Small
viii)	Supra-Orbital ridges	'Trace' or Smooth
ix)	Chin shape	U-shaped

x)	Ramus Flexure	Flared
xi)	Size	Large
xii)	Shape of skull	Broad or Round
xiii)	Mandible	Large and Very prominent.

Que- 13.1 (5)

Ans Darwin tubercle defined the ear lobes are different from individual to individual. Some anomalies are found at the time of the birth which helps in the personal identification.

In India, it was followed in the 1892 but then the

the fingerprints are more reliable than the ear lobes in the personal identification.

Scar Marks in personal identification.

Scar marks are the marks that can be done by the any sharp thing like knife, from falling, etc.

Scar marks are temporary in nature

It helps in the personal identification that who was that person from the Scar marks by their family members.

Occupational Marks -

- Occupational marks are the marks that can be done by their work.
- For an example. A person in the factory so on their body there are occupational marks on present so it helps in personel identification and know that the backgrounds from their occupational marks.

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- Occupational marks are permanent in nature which stays for a long time or whole life.

So, their are some forensic importance of darwin's tubercle, ear lobes, supra-orbital ridges, scar marks or occupational mar which helps in personal identificat

(45)

Que - 14.1

Ans - Portrait Parle / Bertillon System -

Portrait Parle is also known as 'speaking likeness'.

- It is focused on the subject for the virtual examination.

- Alphonse Bertillon was the person who describe the portrait parle method in 1880.

- He tells that portrait parle helps to identify the criminal by their photo.

- He always clicks the photo of the criminal like -

- A Front View
- A Side View

Which helps to identify the characteristic features of the Criminal and write on the diary.

- For example - If when a Crime happened and eye - witness saw the face of the suspect then eye - witness tell the look of the suspect to the bertillon which helps to identify the person.

- But after some time biometric methods are more reliable than the portrait parle method.

Strength of Portrait Parle Method

- Helps to identify the suspect easily.
- Helps in the forensic investigation to solve case
- Reliable and give some positive characteristic
- Identify the person by the sketching given from the witness.

Weakness -

- Biometric Methods are more reliable than Portrait
- Biometric are permanent but portrait not.
- Easily available fingerprint marks but portrait not
- Solve the case easily by the fingerprint than portrait.

Que - 15.2

Ans - Facial Reconstruction. It is a Method to reconstruct the face of an individual from their skeletal remains.

There are three techniques used in forensic anthropology.

i) Facial Reconstruction
ii) Facial Restoration
iii) Skull / Photographic Superimposition.

- Skull / Photographic Superimposition / Video Superimposition

In this technique two photos are taken and ~~comp~~ superimposed on each other. One photo is taken at the front of the skull.

And other is taken at the downward of the either side. When the photo are superimposed So it conclude that it is the same person.

9) Facial Restortation - In this method, Reconstruct the facial features of an individual. Firstly take the photo on the skull, and markings the point to restore the facial. But it is not more reliable because it is not necessary that the skull is found in proper way.

(3) Facial Reconstruction. In this method, Reconstruct the facial features of an individual using the methods like clay, paper. etc. which helps to identify the person. It gives proper feature on the skull and it is also most helpful.

Manual methods. Manual method are the method which can be done by the method like sketching,

For example. Firstly take the paper and put on the skull then with the help of the marker sketch the features. After marking them is a sketch of the skull and then by it identify the features.

Section - C

Que - 17.

Ans

	Features	Male skull	Female skull
1)	General appearance	Large	Small
2)	Architecture	Rugged	Smooth
3)	Orbits	Round with sharp Margins	Small with large Margins
4)	Gonial angle	Angled	Vertical
5)	Cheek shape	Broad	Narrow
6)	Robustiity	High pronounced	Less pronounced
7)	Frontal eminencies	Small	Large
8)	Supra-orbital ridges	'Thick' or smooth	Slight or Vertical

9)	Ramus Flexure	Flared.	Less prominent
10)	Size	Large	Small
11)	Shape of the skull	Broad or Round	Narrow or Vertical

The morphology characteristics of the male skull and female skull are helpful in forensic Anthropology.

They helps to differentiate that it is male skull or female skull.

From the morphological feature forensic investigators also determine the sex from the individual.

For an example - The male skull is very prominent and large. In case of mass disasters Forensic investigators found that it is male because their skull is large compare to the female skull.

If the skull is less prominent and small it is conclude that it is female.

Also examine the different characteristics features of the skull which are different from the male and female skull to for identification.

Que - 16.2

Ans

Human skeletal system

In Human skeleton system there are 206 bones are present.

They are divided into two parts

- Axial skeleton
- Appendicular skeleton

Axial skeleton - Axial skeleton is defined as the center core of the axis. In axial skeleton there are 80 bones are present.

In the skull there are 22 bones are present.

- Frontal 1
- Temporal (2)
- Parietal (2)
- Ethmoid 1
- Mastoid 1
- Occipital 1

(2) In the Vertebral Column there are 24 bones.

(7) Coccyx
(4) Sternum
(5) Ribs

Appendicular skeleton - In the appendicular skeleton there are 126 bones are present.

- Upper Limb
- Lower Limb
- Pelvic Girdle
- Shoulder Girdle

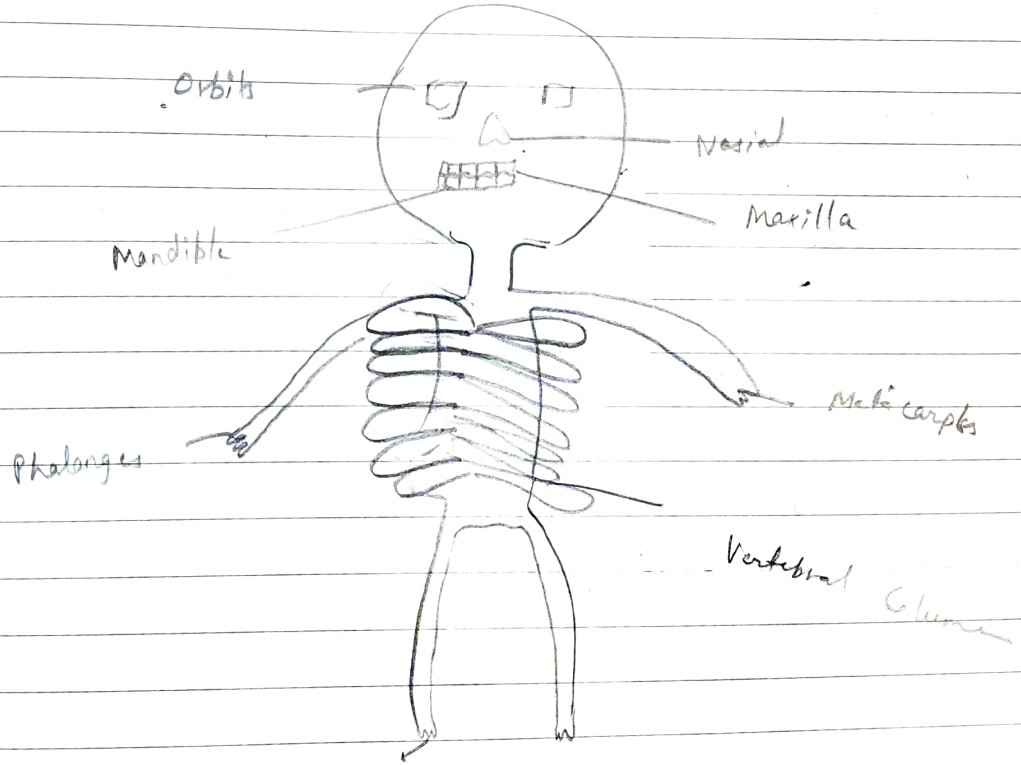


Fig- Skeleton

Que - 18.1

Ans - Somatometric measurements

(1) Head (g-op) - It is the measurement from the glabella to the opithocranion

Glabella - Glabella is present is the front of the forehead.

- It is present b/w the two eyebrows.

Opithocranion - It is present at the side of head.

It is present in the middle of head of the back side.

(2) Face - It is the measurement b/w the acronion to the opisthocranion

(3) Nose - (n-sn) - It is the measurement ~~b/w~~ From the nasion and subnasion.

Nasion - Nasion is the present at the upper point of nose below the eyebrows.

Subnasion - Subnasion are present on the tip of nose.

(4) ~~Ear~~ Height - Height of a person Can take From the following steps.

(1) Stand the person straight (90°) angle with barefoot.

(2) Then ask him to stand with the wall with barefoot and toes touch to the wall.

(3) Hand are released and placed in front of the thighs.

(4) Face are straight at the 90°.

(5) Then, take the measurement from the toes to the end of the head.

Instriment - Measuring tape.

(5) Weight of the person -

(1) Ask the person to stand on the weighing machine with barefoot.

(2) There hands are not touch to

The Weighting machine.

① Weight the person with minimal clothes and barefoot.

② Ask the person to stand straight.

③ Before measuring the weight, the scale would be 0.

④ After 0 ask the person to stand on it and when the scale is stop with down the weight.

Instrument. Weighing machine.

Somato metric measurements (head, face, nose, height, weight) in terms of their reliability and accuracy is helpful in the personal identification.

Every person has different characteristics and also their weight and height different.

By the somatometric measurements forensic investigators find the person individual on the basis of their physical characteristics for the personal identification.

(8)

Que. 19.1

Ans - Strength of Facial Superimposition techniques.

(1) Facial Superimposition technique helps to find out missing persons from their skeletal remains.

(2) By the Superimposition technique, forensic investigators try to draw the face of individual.

(3) It is helpful when no other evidence is found.

(4) It takes some time but give accurate results if the skull ~~is~~ in a proper way.

(5) It also help in reconstruct the Scene and identify the individual.

Limitations -

1. It takes more time.
2. If the skull is not found in a proper way it Cannot give accurate measurements.
3. It is less reliable.
4. In the Case of the mass disaster it Cannot be used because the superimposition technique gives positive it Can not results.
5. During examination, mistakes may happen like not proper marking, Computer function not work properly

Que-20.1

Ans Role of Forensic odontology in mass disaster Cases is Very Good method to identify the person on the basis of dental records

- If the teeth are missing it helps to identify the victim.
- Spacing present between the teeth.
- Size of teeth.
- Any anomalies like Caps, are present.
- Also determine the age from the teeth.
- Also know the Sex.

- Dental records are unique to each individual.
- Not everyone teeth are same.
- Teeth are the main evidence when no other evidence is found.
- It gives information of the individual.
- Forensic odontologists can determine the accurate identification from the teeth.

There are different variations present in the teeth

Ok.

- spacing
- Size
- Missing of teeth etc.

It gives the personal and unique identification.

End of Writing

Ready to pass